

Q & A Part 1

Brainster Web Development Academy



Exercise 1

1. Write a JavaScript function that takes a text argument and alerts: "Hello, [name]!".
2. Ask the user to enter a name from the prompt
3. Rewrite the function by using a fat arrow function



Solution

3

```
const name = prompt("Enter your name:");
```

```
const printName = function (firstname) {  
  console.log(` Hello, ${firstname}!` );  
};
```

```
function printNameNamedFunction(firstname) {  
  console.log(` Hello, ${firstname}!` );  
};
```

```
const printNameWithFatArrow = (firstname) => console.log(` Hello, ${firstname}!` );  
printName(name);
```



Exercise 2

1. Create an HTML page with a button.
2. Write a JavaScript function that changes the background color of the page to a random color when the button is clicked.

3

(**Hint:** Use event listeners to handle the button click.)

(**Hint:** Use `Math.random` to generate random color)

(**Tip:** All the colors can be represented by the values in this array:

`['1','2','3','4','5','6','7','8','9','0','A','B','C','D','E','F']`)



Solution

3

```
let button = document.querySelector("button");
let color = prompt("Enter a color");

function changeBackgroundColor() {
  let colorItems = "1234567890ABCDEF"; // or ["1","2"...."A","F"];
  let color = ["#"]; // '#000000'

  for (i = 0; i < 6; i++) {
    let index = Math.floor(Math.random() * 16) + 0;
    let colorRandomNumber = Array.from(colorItems)[index];
    color.push(colorRandomNumber); // ["#", "2"] // ["#", "2", "6"] // ["#", "2", "6", "D"]
  }
  document.querySelector("body").style.backgroundColor = color.join(""); // ["#", "2", "6", "D", "E", "1", "2"] => #26DE12;
}

button.addEventListener("click", changeBackgroundColor);
```



Exercise 3

1. Create an HTML page with a list of numbers (e.g., `<ul>` with `<li>` elements) and a button.
2. Write a JavaScript function that finds the largest number in the list and displays an alert with that number when the button is clicked.



Solution 1

3

```
let listItems = document.querySelectorAll("#numbers li");

function findMax() {
  let max = +listItems[0].textContent;
  listItems.forEach((element) => {
    const value = +element.textContent;
    if (value > max) {
      max = value;
    }
  });
  return max;
}

let maxValue = findMax();
console.log(maxValue);
```



Solution 2

3

```
let listItems = document.querySelectorAll("#numbers li");

function findMax() {
  let arrayOfNumbers = [...listItems].map((element) => Number(element.textContent));
  let arrayOfNumbers = Array.from(listItems).map((element) =>
    Number(element.textContent)
  );
  return Math.max(...arrayOfNumbers);
}

let maxValue = findMax();
console.log(maxValue);
```



Exercise 4

1. Create an HTML page with a list of numbers (e.g., `<ul>` with `<li>` elements) and a button.
2. Write a JavaScript function that removes duplicate elements from the array and console logs the new array with unique numbers.



Solution

3

```
const listItems = document.querySelectorAll("#numbers li");

function findDuplicates() {
  const arrayFromListItems = Array.from(listItems); // [...listItems] => [li, li, li, li, li, li, li, li] as Array in order to use map
  const arrayFromNumbers = arrayFromListItems.map((el) => +el.textContent); // [55, 5, 55, 26, 26, 4, 4, 44, 11]

  return arrayFromNumbers.filter((element, index, array) => {
    return array.indexOf(element) === index;
  });
}

const result = findDuplicates();
console.log(result);
```

